Probability & Stochastic Processes

Rigorous foundations to computational probability

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Abstract

We will cover probability from rigorous foundations to stochastic calculus, ergodic theory, Monte Carlo, filtering, and sampling.

Contents

1 PART I — FOUNDATIONS

1.1 1. Probability Spaces and Random Variables

- Probability spaces; σ-algebras; Kolmogorov axioms
- Random variables; distribution functions; pushforward measures
- Expectation via Lebesgue integration; moment generating functions
- Independence; product spaces; Kolmogorov extension theorem

1.2 2. Classical Limit Theorems

- Laws of large numbers: weak (Chebyshev) and strong (Borel-Cantelli)
- Characteristic functions; Lévy continuity theorem
- Central limit theorem: Lindeberg-Feller condition; Berry-Esseen rates
- Infinitely divisible distributions; stable laws (brief)

1.3 3. Conditional Expectation and Martingales

- Conditional expectation as L^2 projection; properties; tower property
- Discrete-time martingales; stopping times; filtrations
- Optional stopping theorem; Doob's inequalities (maximal, L^2, upcrossing)
- Convergence theorems: martingale convergence; uniform integrability

1.4 4. Markov Chains (Discrete Time)

- Transition kernels; Chapman-Kolmogorov equations
- Recurrence and transience; hitting times; classification of states
- Stationary distributions; detailed balance; ergodic theorem
- Mixing times; spectral gap; connection to spectral graph theory (Course 2)

2 PART II — CONTINUOUS-TIME PROCESSES

2.1 5. Brownian Motion

- Construction: Lévy-Ciesielski and Kolmogorov approaches
- Path properties: continuity, nowhere differentiability, quadratic variation
- Strong Markov property; reflection principle; hitting times
- Connections to harmonic functions; Kakutani's theorem

2.2 6. Stochastic Calculus

- Itô integral: L^2 construction; isometry property
- Itô's formula: single and multi-dimensional
- SDEs: existence and uniqueness (Lipschitz coefficients)
- Girsanov theorem; Novikov condition; Cameron-Martin

2.3 7. Diffusion Processes and Generators

- Generator of a diffusion; Kolmogorov forward (Fokker-Planck) and backward equations
- Stationary distributions; conditions for existence and uniqueness
- Connections to semigroup theory; Feynman-Kac formula
- Diffusions on manifolds (brief)

2.4 8. Lévy Processes and Jump Processes

- Characteristic exponent; Lévy-Khintchine representation
- Poisson random measures; compensation
- Stochastic integrals with jumps; Itô formula with jumps
- Applications: finance; insurance ruin; queueing

2.5 9. Gaussian Processes

- Covariance functions; mean functions; RKHS connection
- Sample path regularity via Kolmogorov-Chentsov
- Gaussian process regression; posterior computation; marginal likelihood
- Applications: Bayesian optimization; probabilistic numerics (Course 4)

3 PART III — ERGODIC THEORY AND MIXING

3.1 10. Ergodic Theory

- Measure-preserving systems; ergodicity; invariant measures
- Von Neumann L^2 ergodic theorem
- Birkhoff pointwise ergodic theorem
- Mixing conditions; mixing for Markov chains and diffusions
- Applications: time series; statistical mechanics; MCMC convergence

3.2 11. Convergence Rates and Mixing

- Total variation distance; coupling inequality
- Poincaré inequality; spectral gap; Cheeger inequality connection
- Log-Sobolev inequalities; hypercontractivity; Bakry-Émery criterion
- Coupling methods; maximal coupling; Lindvall's coupling

4 PART IV — COMPUTATIONAL PROBABILITY

4.1 12. Monte Carlo Methods

- Classical Monte Carlo; variance reduction (control variates, stratification, antithetics)
- Metropolis-Hastings; Gibbs sampler; detailed balance
- Hamiltonian Monte Carlo; MALA; NUTS
- Convergence diagnostics; effective sample size; Gelman-Rubin

4.2 13. Sequential Monte Carlo and Filtering

- Particle filters; sequential importance resampling; degeneracy
- Kalman filter; Kalman-Bucy filter; derivation and optimality
- Nonlinear filtering; Zakai equation; Kushner-Stratonovich
- Applications: robotics (SLAM), weather forecasting, finance

4.3 14. Large Deviations for Processes

- Cramér's theorem on path spaces; Schilder's theorem for BM
- Freidlin-Wentzell theory: quasi-potential; rare events for diffusions
- Applications: MCMC efficiency; transition state theory; rare event simulation

4.4 15. Stochastic Optimization and Sampling

- Langevin dynamics; overdamped vs. underdamped; connections to MCMC
- SGD as stochastic differential equation; continuous-time limit
- Stein variational gradient descent
- Interacting particle systems; propagation of chaos; McKean-Vlasov limits